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④ Construct a Huffman code for

Symbol	A	B	C	D	E
Freq	0.33	0.32	0.03	0.17	0.15
	②	④	①	③	⑤

Step 1 Arranging in terms of freq

0.03	0.15	0.17	0.32	0.33
C	E	D	B	A

Step 2 Combining the smallest nodes

$C+E$	$D+(CE)$	$B+A$
0.18	0.35	0.65

$$DCE + BA = 0.65 + 0.35 = 1$$

Decode

i) ~~10010110010101~~ using the code of question

Encode ABACABAD

A = 11, B = 10, C = 0, D = 00

Encoding it $\Rightarrow$ 11/10/11/0/11/10/11/10

Hence = 1110110101110110

ii)

10010110010101
B C A B C
= BCABC